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A new pilot absorber for CO_2 capture from flue gases: Measuring and modelling capture with MEA solution

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ABSTRACT

A pilot absorber column for CO_2 recovery from flue gases was constructed and tested with aqueous 30 wt% monoethanolamine (MEA), a primary amine, as capture solvent. The pilot plant data were compared with a mathematical rate based packed-column model. The simulation results compared well with the pilot plant data.

The packed height of the column can be varied from 1.6 to 8.2 m by means of five different liquid inlets. The column has an inner diameter of 100 mm and is packed with structured Mellapak 250Y packing. Counter-current flow is used.

The pilot plant performance was investigated by changing three parameters: the absorption height, liquid flow rate, and the loading of lean MEA. This was done using a synthetic flue gas consisting of 10% CO_2 with a flow rate of approximately 33 m^3/h at ambient temperature and atmospheric pressure. 23 runs were performed.

It was observed that while CO_2 recovery increases with an increase in flow rate of absorbent and absorption height, it decreases as the lean CO_2-loading of the absorbent increases. In addition it has been possible to obtain temperature bulges in the bottom part of the absorber by the applied operation conditions. Bulges are observed at liquid flows around 4.2 L/min and below. The results showed that is was possible to achieve 80% recovery with 3.3 m absorption height and a liquid flow of 2.1 L/min. The simulations show good agreement with the experimental values, although slight deviations arise as the CO_2-loading increases and the temperature bulge becomes more distinct.

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1. Introduction

Carbon dioxide is emitted from large sources of fossil fuel combustion, particularly power plants. It is one of the key man-made greenhouse gases that must be minimized in order to avoid catastrophic climate change. The capturing of trillions tons of CO_2 is a daunting task, but is a challenge that has to be overcome successfully if disaster is to be averted (Schrag, 2007).

The process for post combustion capture consists of four stages as sketched in Fig. 1. First, CO_2 is removed from the flue gas by absorption in a packed absorber column. Then 'rich' solvent containing CO_2 is led to a desorber column and heated in a reboiler to release the CO_2. Next, CO_2 is compressed and transported to a geological storage site or injected into an oil and gas reservoir. Finally the regenerated 'lean' solvent is recycled to the absorber.

The supply of heat for solvent regeneration is by far the most expensive aspect of the post combustion capture process. The cost of solvents, electricity, energy for pumps, and cooling are also high but to a lesser extent (Rao and Rubin, 2002).

A number of studies have been carried out in order to investigate CO_2 capture in pilot plants using MEA. Most of these have appeared since 2006, with the exception of a few earlier studies. The older work conducted at University of Regina in Canada is described by Tontiwachwuthikul et al. (1992), Aroonwilas et al. (1999), and Aboundheir (2002) and more recently by Idem et al. (2006, 2009). There is a growing activity on post combustion capture in Norway evidence by the MEA related studies of Tobiesen et al. (2007) and Hessen (2010). Work has been performed at University of Kaiserslautern (Notz et al., 2007) as well as several studies at University of Texas, Austin (Dugas, 2006; Kvamsdal and Rochelle, 2008). A number of larger industrial funded consortium projects have also appeared (Hessen, 2010; Dugas et al., 2009; Lee et al., 2008).

While some studies have been performed concerning the mechanisms of CO_2 capture from flue gases using solvents, there is limited pilot plant data for such systems, where enough information has been reported to enable model development. Here we report the construction and validation of a new pilot-scale facility, which we hope will contribute to research in CO_2 capture, as this

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Nomenclature

a	specific wetted area for mass transfer, m^2/m^3
A_c	cross sectional area of the column, m^2
$c_{p,i}$	molar heat capacity of component in the gas phase, J/(mol K)
$c_{p,G}$	molar heat capacity of the gas, J/(mol K)
$c_{p,L}$	molar heat capacity of the liquid, J/(mol K)
E	enhancement factor
E_∞	enhancement factor for instantaneous reaction
F_{SE}	factor for surface enhancement
G	molar gas flow, mol/s
H_{CO_2}	Henry's law constant for CO_2, J/(s K m^2)
h	heat transfer coefficient in gas, J/(s K m^2)
ΔH_{H_2O}	heat of condensation of H_2O, J/mol
ΔH_{CO_2}	heat of condensation of CO_2, J/mol CO_2
$K_{G,i}$	overall gas side mass-transfer coefficient of component i, mol/(m s Pa)
$k_{G,i}$	gas side mass-transfer coefficient of component i, m/s
$k_{L,i}$	liquid side mass-transfer coefficient of component i, m/s
L	molar liquid flow, mol/s
M	Ha^2, Ha is the Hatta number
N_i	molar flux of component i, mol/(m^2 s)
p_i	partial pressure of component i in the bulk gas phase, Pa
p_i^*	partial pressure of i component in gas phase if it were in equilibrium with the liquid phase, Pa
q	heat flux, J/(m^2 s)
T_L	temperature of liquid phase, K
T_G	temperature of gas phase, K
x_i	mole fraction of component i in the liquid phase, mol/mol
X_{CO_2}	mole fraction of CO_2 in reacted and unreacted form, mol/mol
y_i	mole fraction in gas phase of component i, mol/mol
z	absorption height, m

Nomenclature for PI diagram

Ci1	CO_2-transmitter, measuring inlet flow, mol/mol
Ci2	CO_2-transmitter, measuring outlet flow, mol/mol
TI1–T10	temperature in column, °C
TI43	temperature in the liquid inlet, °C
FI	flowmeter, L/min
PI	pressure measurement, mbar g

important area moves increasingly to pilot and even demonstration scale.

A vast number of both structured and random (dumped) packing types are available from commercial suppliers. Different packing types have been used in the above-mentioned works. Structured packing was considered in Hessen (2010), including 250Y Mellapak, whereas dumped packing was used by Lee et al. (2008), Dugas et al. (2009), Kvamsdal and Rochelle (2008), and Tontiwachwuthikul et al. (1992). Here the Koch-Glitsch IMTP 25, 40, 50 and ceramic Berl saddles, 12.7 mm, were used. Some studies included the same pilot facility but with different packing material. Aboundheir (2002) tested the A4 structured packing and the two dumped packings, IMTP 15 and Pall Ring-16mm. Dugas (2006) tested the structured Flexipac 1Y and IMTP 40 dumped packing. Finally Aroonwilas et al. (1999) tested the Gempak 4A structured and IMTP dumped packings. Both packings were tested in three different sizes. They found that the structured packings performed better than dumped

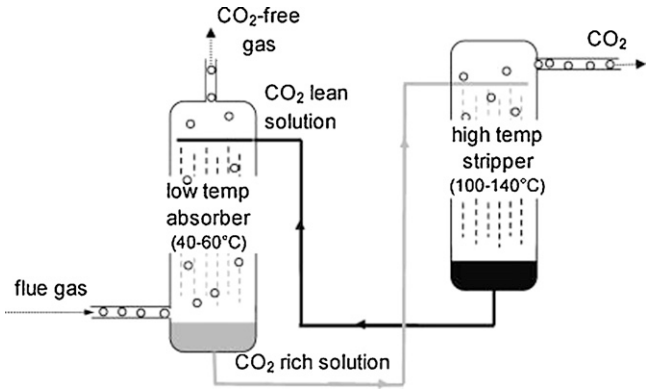


Fig. 1. A simple illustration of a CO_2 capture plant.

packings, because of the greater surface area per unit volume. However there are some disadvantages, as the structured packings require excellent initial distribution to obtain the full benefit. Finally the cost of structured packings is also higher than dumped packings (Wilson, 2004).

While packing suppliers should have a comprehensive understanding of the performance of their packing, far from all packings have received a thorough and independent comparison in the open literature. An interesting study could be performed across research laboratories where different pilot plants independently compared the same packings. On the other hand, having validated our pilot plant here for a given packing and a "standard" solvent (MEA) we have confidence that the same plant could be used to evaluate both different packings and different capture solvents.

Previous pilot scale studies (Tontiwachwuthikul et al., 1992; Aroonwilas et al., 1999; Aboundheir, 2002; Idem et al., 2006, 2009; Tobiesen et al., 2007; Hessen, 2010; Notz et al., 2007; Dugas, 2006; Kvamsdal and Rochelle, 2008; Dugas et al., 2009; Lee et al., 2008) cover a range of operation conditions. Three input variables are interesting from a process point of view. These are the solvent CO_2 loading, solvent concentrations, and flue gas composition. The lean loading covered is 0–0.35 and rich loading is 0.33–0.49 mol_{CO_2}/mol_{MEA}. The concentration of MEA in the MEA-H_2O solvent varies from 18 to 54 wt%. Most of the experiments are carried out using 30 wt% MEA concentration. The CO_2 concentration in the flue gas coming into the bottom of the absorber is in the range of 4–20 vol%.

Several analytical methods have been used in the determination of liquid CO_2 content for loading evaluation. Of the studies above, CO_2 content was mainly found by a barium chloride method very similar to the one outlined in Appendix A of this work. It is cheap but time consuming. Other methods are the Total Organic Carbon (TOC) analysis, used by Dugas (2006) and gas chromatography–mass spectrometry (GC–MS) used by Idem et al. (2009). The benefit of this last method is that it is on-line and unmanned measurements can be performed. This method was not pursued because of the high cost involved. A titration method described by Horwitz (2006) may also be used. We chose to proceed with the Barium Chloride method since we had prior experience with it.

The absorber temperature profile (temperature along the length of the column) can be measured by means of appropriately placed temperature probes. The reaction between CO_2 and the solvent is exothermic, resulting in a steep temperature increase at certain conditions. An observations by Kvamsdal and Rochelle (2008) was that the location of the temperature bulge can move from the bottom to the top of the column. The temperature bulge is quantified by its magnitude and location in the column and is seen as a convex curve on the temperature profile. At a critical liquid to gas (L/G)-ratio the temperature bulge will be in the middle of the

column, where the rich loading can be maximized while maintaining a reasonable CO_2 removal. This phenomenon should be analysed more closely and there is a need to determine effect on capture effectiveness.

The aim of this work is to report results for a new pilot-scale (10 m high, 10 cm internal diameter) absorber. Results are reported for Mellapak 250Y structured packing and MEA solution as capture solvent. In addition to reporting this new data, which can be used for model development and testing (here it is tested against an existing published model (Gabrielsen et al., 2006; Faramarzi et al., 2009)), a secondary aim is to validate the plant for a given packing and solvent in order to proceed to other packings and solvents with greater confidence. The following operation conditions were covered: A lean loading of 0.11–0.30 and a rich loading of 0.15–0.39 mol_{CO_2}/mol_{MEA}. A 30 wt% aqueous MEA solution and a synthetic flue gas composition of 9.2–10.9 vol% CO_2. A modified and optimized barium chloride method was developed for the analysis of the liquid phase for the determination of the total dissolved CO_2 (see Appendix A). The plant was fitted with two storage tanks with a volume of 1 m^3 each. Therefore it was possible to run a series of experiments with the same lean loading and MEA concentration. The basis of the different operation conditions are consequently very similar and therefore comparable. The results of the various experimental runs contain detailed information about the loading and temperature profiles throughout the column. Additionally the CO_2 concentration in the gas phase was measured at both the gas inlet and outlet of the column.

The data were also modelled using a previously developed rate-based model (Gabrielsen et al., 2006; Faramarzi, 2010). While further development of the model was not undertaken as part of this work, the production of new pilot-plant data did provide further validation of the model, as usable pilot plant data are not plentiful.

2. Modelling CO_2 absorption

The experimental pilot runs performed in this work were modelled using a rate based column model originally developed by Gabrielsen et al. (2006, 2007) and Gabrielsen (2009) and further modified by Faramarzi et al. (2009, 2010) and Faramarzi (2010). The main purpose of this work was the experimental validation of the pilot scale absorber. More detail on the model can be found in previous work (Gabrielsen et al., 2006, 2007; Gabrielsen, 2009; Faramarzi et al., 2009, 2010; Faramarzi, 2010) and only the basic elements are presented here. Previous works have presented models for mass transfer coefficient, mass transfer area, hydraulics and physical properties. A comprehensive comparison of the various components that make up a complete rate-based model is considered beyond the scope of this work. The model is mass transfer based and accounts for transport properties, thermodynamics, phase equilibrium and reaction kinetics. The thermodynamic model used is extended UNIQUAC (Faramarzi, 2010). Extended UNIQUAC is used here to model the thermodynamic properties required in the full rate-based model, because it is the model where we have most experience (Gabrielsen et al., 2006, 2007; Gabrielsen, 2009; Faramarzi et al., 2009; Faramarzi, 2010). Any thermodynamic model for electrolytes which is at a reasonable level of development (for example e-NRTL) could have been used, although it was not the purpose of this work to develop and present comparisons of thermodynamic models.

The model is built up around eight differential equations describing the mass and energy balances of the gas and liquid, to which a number of equations are related in order to account for the dependency of different parameters on temperature, concentration, pressure, etc. In the development of the model some general assumptions were made in order to simplify the system sufficiently for modelling:

1 The reaction rate is fast enough so that all the reaction will occur in the liquid film while the CO_2 concentration in the bulk phase is assumed to be in equilibrium.
2 Liquid side heat transfer resistance is small compared to the gas phase, thus the interface temperature is the same as the bulk temperature.
3 The liquid side mass-transfer resistance for the volatile solvent is negligible.
4 The interfacial surface area is the same for heat and mass transfer.
5 The system is adiabatic.

The following equations are used in the description of the basic behaviour of mass and energy:

$$\frac{dG}{dz} = -(N_{CO_2} + N_{H_2O})aA_c \tag{1}$$

$$\frac{dy_{CO_2}}{dz} = \frac{N_{CO_2}aAc(y_{CO_2} - 1) + N_{H_2O}y_{CO_2}aA_c}{G} \tag{2}$$

$$\frac{dy_{H_2O}}{dz} = \frac{N_{H_2O}aAc(y_{H_2O} - 1) + N_{CO_2}y_{H_2O}aA_c}{G} \tag{3}$$

$$\frac{dL}{dz} = -N_{H_2O}aA_c \tag{4}$$

$$\frac{dX_{CO_2}}{dz} = \frac{(N_{H_2O}X_{CO_2} - N_{CO_2})aA_c}{L} \tag{5}$$

$$\frac{X_{H_2O}}{dz} = \frac{N_{H_2O}(X_{H_2O} - 1)aA_c}{L} \tag{6}$$

$$\frac{dT_G}{dz} = -\frac{qaA_c}{Gc_{p,G}} \tag{7}$$

$$\frac{dT_L}{dz} = \frac{(N_{CO_2}c_{p,CO_2} + N_{H_2O}c_{p,H_2O})aAc(T_L - T_G)}{Lc_{p,L}} - \frac{qaA_c}{Lc_{p,L}} + \frac{(N_{CO_2}\Delta H_{CO_2} + N_{H_2O}\Delta H_{H_2O})aA_c}{Lc_{p,L}} \tag{8}$$

G and L describe the total flow in moles per minute in respectively the gas and liquid phase while y_i is the mole fraction of component i in the gas phase. X_i denotes the mole fraction in the liquid phase based on a CO_2 free environment, and thus the liquid L is also on a CO_2 free basis. T is the temperature in kelvin and a is the wetted area in the column and A_c is the cross sectional area of the column. z is the position in the column while $c_{p,i}$ is the heat capacity of component i. The fluxes, N, are given in terms of mass transfer coefficients and driving forces of partial pressures, and for CO_2 can be written as:

$$N_{CO_2} = K_{G,CO_2}(p_{CO_2} - p^*_{CO_2}) \tag{9}$$

$$\frac{1}{K_{G,CO_2}} = \frac{1}{k_{G,CO_2}} + \frac{H_{CO_2}}{Ek_{L,CO_2}} \tag{10}$$

where the mass-transfer coefficient k_{G,CO_2} of Rocha et al. (1996) is used.

The heat flux between the gas and the liquid phase can be expressed as:

$$q = h(T_G - T_L) \tag{11}$$

Eqs. (1)–(3) represent the mass balances of the gas phase. Eqs. (4)–(6) describe the mass balance of the liquid phase. Finally, Eqs. (7) and (8) are the energy balances used in order to calculate the temperatures of the phases.

Table 1
Enhancement factor according to different theories (Faramarzi et al., 2010).

van Krevelen and Hoftijzer (1948)	$E = \frac{\sqrt{M((E_\infty - E)/(E_\infty - 1))}}{\tanh\left(\sqrt{M((E_\infty - E)/(E_\infty - 1))}\right)}$
Astarita et al. (1983)	$E = \left[\left(1 + \frac{n^{n/n-1}(E_\infty - 1)^{1/n-1}E_\infty}{(1+M)^{n/2n-2}}\right) - 1\right]^{1/n} \cdot \frac{(1+M)^{n/2n-2}}{n^{n/n-1}(E_\infty - 1)^{1/n-1}}$
Pseudo-first order (Kucka et al., 2003)	$E = \sqrt{M}$

The thermodynamic component of the model is used in the calculation of gas and liquid properties in order to estimate heat of CO_2 absorption, heat capacities, Henry's constants, and partial pressures (which are required to provide an estimate of various driving forces). Billet and Schultes (1999) and Rocha et al. (1993, 1996) have provided equations for mass transfer coefficients.

In this study the correlation by Rocha et al. was used, since it gave better agreement between the modelling and experiments performed in this work. Correlation parameters were obtained from the original work of Rocha et al. The surface enhancement factor (F_{SE}) was set equal to $F_{SE} = 0.25$ (in Rocha et al., 1996, it is set to 0.35). The reason for the deviation from the Rocha et al. correlation is that the pilot plant used in this work does not have as high a surface efficiency of an industrial sized column which has a much greater diameter.

The kinetic component of the model is used in the calculation of the enhancement factor, E, in Eq. (10), where it describes the excess driving force obtained due to reaction between the solvent and CO_2 compared to the case without reaction. An investigation of the enhancement factor theory was made by Faramarzi et al. (2010) where a discussion is given of the more advanced implicit and simple explicit enhancement factor equations. It was concluded that all enhancement factors gave good results for the absorption part. In this work, the more advanced implicit equation derived by Krevelen and Hoftijzer was used. It is a more general equation which can also be used in the stripper calculation. In the enhancement factor equation derived by Astarita et al. (1983) (see Table 1) an explicit expression has been derived. It involves a variable n which is a constant describing the smoothness of the transition of CO_2 from gas phase to the liquid phase. A higher value of n gives a smoother transition. The square root of M is the Hatta number and E_∞ is the enhancement factor for instantaneous reaction. For the conditions of the experiments performed here, all enhancement factors gave reasonable agreement with experiments, although it may be expected that at other conditions (for example in the stripper) the pseudo first order approximation may no longer be valid.

The rate-based model is solved as a boundary value problem (BVP) involving Eqs. (1)–(8). The boundary conditions are the defined inlet flows of flue gas, liquid composition, pressure, and temperature. The main results are the properties of the cleaned gas and rich liquid. The model is thus capable of predicting more than just the temperature and the CO_2-loading profiles: Compositions, flows, transport properties, thermodynamic properties, and kinetic properties throughout the length of the column are also calculated.

3. Experimental setup

The pilot plant described in this work is shown in Fig. 2. The absorber column has a total height of 10.5 m and an internal diameter of 0.1 m. It is made of pyrex glass and packed with Sulzer Mellapak 250Y. The column consists of 10 sections each 1.0 m high. Redistributors are located between sections. The packing height for each section is 0.82 m and thus a maximum effective absorption height of 8.2 m can be obtained. The packing diameter is 0.084 m. Liquid can be introduced every second section, i.e. every two meters. This is done in order to vary the effective height of the packing. The untreated gas feed is introduced at the bottom of the column. The column operates in counter-current mode – gas flowing upward contacts liquid flowing down the column.

Liquid samples can be taken from every section using custom-made sampling valves, V1–V10 in Fig. 2. These consist of special liquid collection rings which are machined into the section connectors. Samples can also be taken from the storage tanks T1 and T2.

Pt100 temperature probes are inserted between every section. Additional temperature probes are placed at the inlets and outlets and in the recycled cleaned gas phase. Three pressure transmitters from Bugh & Bønsøe – the Tensto 6300 series – are placed in the bottom, middle, and top of the column, as indicated by PI1, PI6, and PI11.

The gas phase composition is measured at the inlet and outlet of the column, Ci1 and Ci2 in Fig. 2. The carbon dioxide transmitters used are Vaisala CARBOCAP® series GMT221. These cover a range of 0–20% CO_2, with a ±2% expected accuracy of the full scale reading. Flow is measured manually by four calibrated rotameters FI1–FI4.

In the top of the absorber unit there is a demister with a subsequent steam trap to ensure that no solvent is recycled with the gas stream. It is possible to cool down the recycle gas with heat exchanger HE1, although this was not used for these experiments.

Two storage tanks each with a volume of 1000 L are located next to the column. Before a run commences stock solution of the desired composition is prepared and stored in one of the tanks. The storage tanks are large enough to collect liquid from several runs. This gives the possibility to perform similar experiments with the same concentration of MEA and lean loading of the solution. One storage tank holds the spent rich solvent while the other contains fresh solvent with a known loading.

3.1. Chemicals used

85 wt% (99.8% purity) MEA was obtained from BRENNTAG Nordic, Denmark. The solvent was diluted using demineralized water to the desired concentration of 30 wt% MEA. A synthetic flue gas was created by nitrogen and carbon dioxide from gas cylinders from AGA Denmark, with purities of 99.95% and 99.8% respectively. For the liquid analysis of CO_2 using the barium chloride method, 0.1 M NaOH and 0.5 M HCl solution were obtained from VWR. The concentrations are known with an accuracy of 0.1%. The 0.5 M Barium Chloride solution was prepared from $BaCl_2 \cdot 2H_2O(s)$ from VWR dissolved in water.

4. Experimental procedure

Before a run begins, pure nitrogen is continuously recycled in the column in order to minimize degradation of the solvent from air oxidation. The lean liquid flow valve is then opened and CO_2 flow adjusted in order to reach a defined inlet gas concentration. The recycle gas flow is kept constant in all experiments in the rage 470–540 L/min. The CO_2 concentration was maintained at 10%. The CO_2 flow varied from 107 to 113 g/min, depending on the overall flow of gas. The concentration of MEA was 30 wt% based on the initial batch mixed in the storage tank. The lean MEA was at room temperature varying between 19.5 and 23 °C. The atmospheric pressure in the pilot plant hall was between 1.000 and

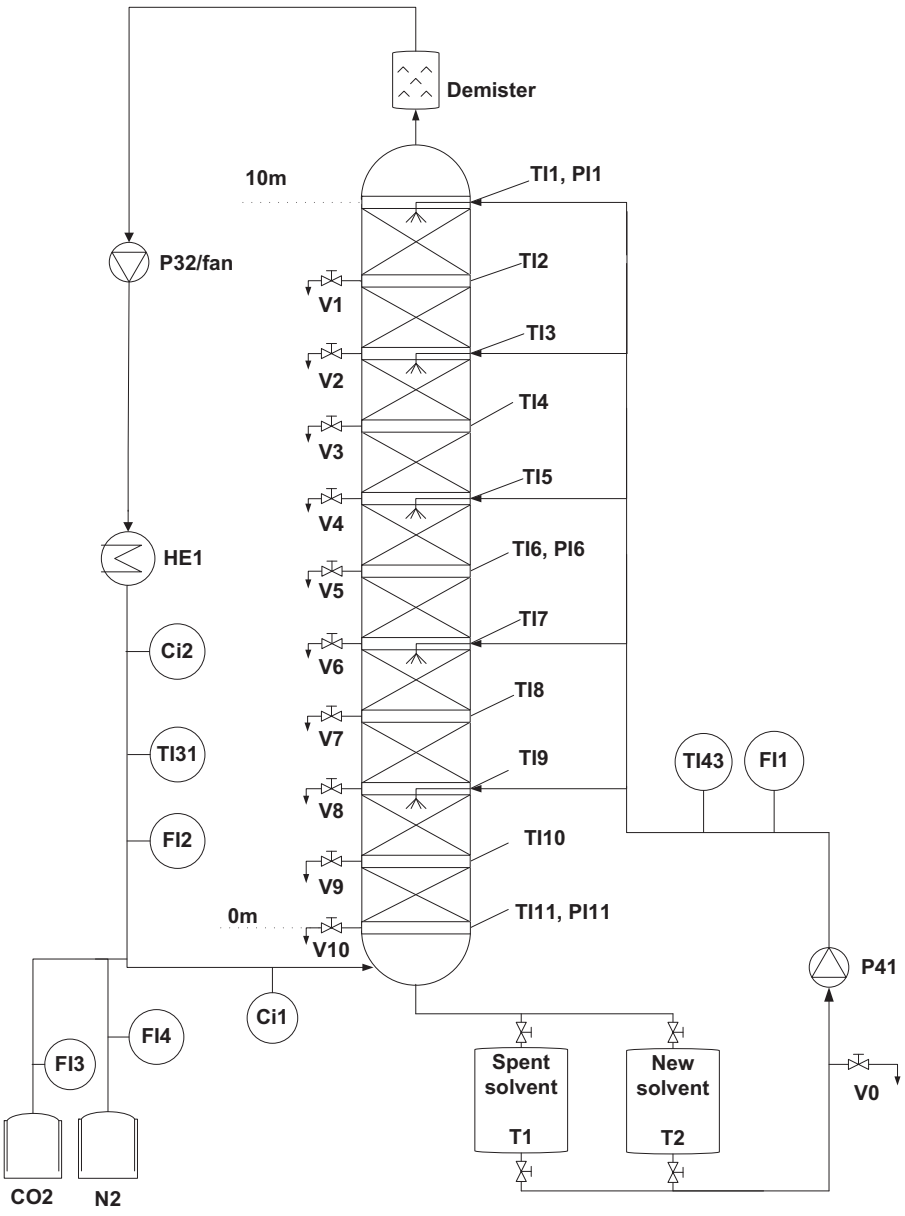


Fig. 2. Simplified piping and instrumentation diagram of pilot scale CO_2 absorber unit.

1.017 bar. With these parameters held relatively constant, the aim was to investigate the effect on the process of the liquid flow rate, the height of absorption, and the variation of lean loading of CO_2.

4.1. Obtaining steady state

Steady state was usually obtained within 5–25 min, depending on the operation condition and previous set point. Steady state was assumed to have been attained when the internal column temperatures and the inlet and outlet CO_2 concentrations deviated from their final values by less than 4% (see Fig. 3). A process variability limit within 4% was felt to be an adequate representation of steady-state.

4.2. Sampling

After reaching steady state, data and liquid samples were collected within 5 min and subsequently analysed. A slight positive pressure is needed inside the column in order to extract samples from valves V1 to V10. To make sure that this can be achieved, 0.7 g/min N_2 is added to the gas stream continuously in all experiments, an amount small enough not to have any impact on the experiments.

4.3. Analysis of samples

The CO_2 content in the liquid sample was determined by the so-called the barium chloride method (Sun et al., 2005). The CO_2 is precipitated as $BaCO_3$, which is dissolved by adding a known amount of 0.5 M HCl and back titrated using 0.1 M NaOH. The amount of CO_2 can then be determined as the difference of excess HCl compared to NaOH. Every sample was analysed twice and the experimental error was estimated to be less than ±4%. To reach this low deviation a great deal of optimization had to be done with respect to the existing procedure. The optimized analysis procedure can be found in Appendix A.

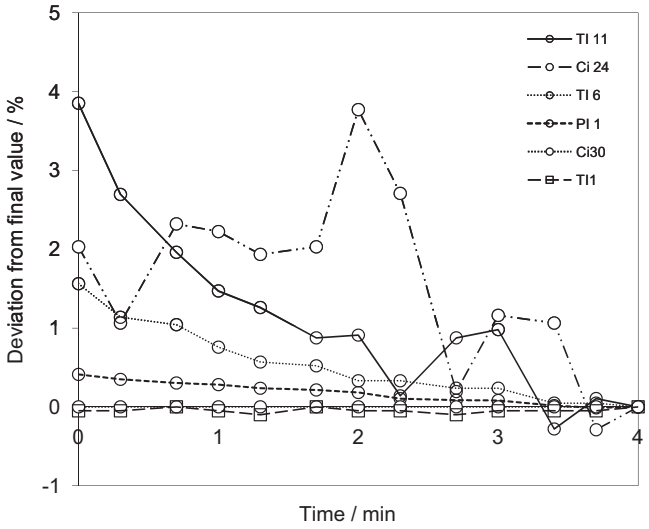


Fig. 3. Steady state values for run R5 based on percentage deviation with respect to the value at the end of the experiment.

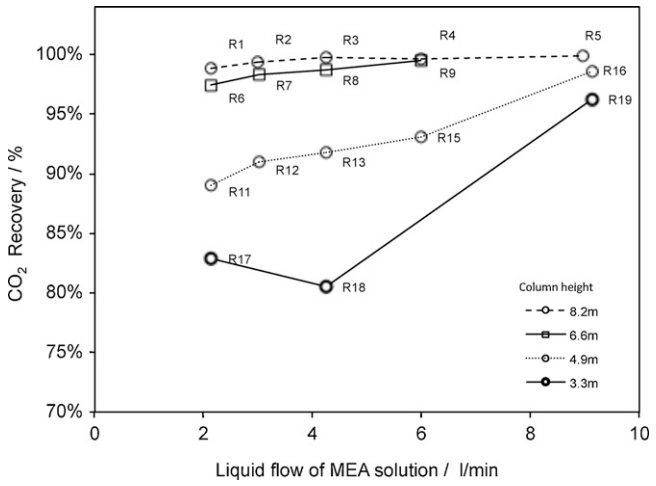


Fig. 4. CO_2 absorption efficiency as a function of liquid flow of MEA varying from 2 to 9 L/min. The gas flow and concentration are kept constant at approximately 510 L/min and 9.8% CO_2 respectively.

5. Results and discussion

A total of 23 steady state experiments were carried out, numbered R1–R23. The runs are divided into two main categories: runs R1–R20 where absorption height and L/G ratio are varied, and runs R21–R23 where lean liquid CO_2 loading is varied. All other process variables were kept constant. An overview of the runs is given in Table 2. For runs R21–R23 both L/G ratio and absorption height were kept constant while lean loading was varied between 0.25 and 0.30 mol/mol.

The results are summarized in Tables 3–6. They depict the operating conditions and measured properties from the absorber bottom to the top. Table 3 shows flows and mass balances, temperature profiles are shown in Table 4, pressure profiles in Table 5, and loading profiles in Table 6. CO_2-loading profiles are given in Table 6 for selected runs.

5.1. Accuracy of the steady state mass balances

A steady state was assumed to exist when process variables varied by less than 4%. In order to ascertain whether steady state was attained, CO_2 balances were set up to determine if all CO_2 was accounted for.

The amount of CO_2 into the system, $n_{CO_2,in}$, is calculated by summing all the streams that are entering the system and multiplying by the mole fraction of CO_2, $y_{CO_2,1}$, measured by the CO_2-transmitter Ci1. Mole fractions of CO_2 measured by CO_2-transmitters can be found in Table 3. The streams included in the total inlet flow ($G_{total,in}$) are: FI2, FI3 and FI4. Besides this contribution there is also some CO_2 in the lean MEA which must be accounted for. The contribution for this lean loading is measured in the lab (Table 6) and the number of moles CO_2 into the system is thus calculated as:

$$n_{CO_2,in} = \sum G_{total,in} \cdot y_{CO_2,1} + n_{CO_2,liquid\,in} \quad (12)$$

The total amount of CO_2 leaving ($n_{CO_2,out}$) can be calculated in a similar way, from the two outlet streams, FI2 and the CO_2 leaving in the rich loading. The mole fraction in the outlet gas ($y_{CO_2,2}$) is measured by CO_2-transmitter Ci2. The expression then becomes:

$$n_{CO_2,out} = G_{total,out} \cdot y_{CO_2,2} + n_{CO_2,liquid\,out} \quad (13)$$

The deviation of the mass balance can now be calculated as the difference of moles CO_2 between inlet and outlet streams divided by the total inlet of CO_2 as:

$$Deviation\,\% = \frac{n_{CO_2,out} - n_{CO_2,in}}{n_{CO_2,in}} \quad (14)$$

A maximum deviation of ±10% is obtained for runs R1 and R12, while the deviation averaged over all runs was 6.4%. Taking the number of measurements into account a certain deviation in mass balance is to be expected. This error could be minimized by installing flow- and temperature transmitters in the flue gas stream inlet at the bottom of the column, thereby reducing the number of measurements for the inlet steams. It should be noted that a lean loading greater than zero will generally result in greater errors, because of the errors inherent in the measurement of CO_2 concentration.

5.2. Recoveries of CO_2 obtained from experiments

An overview of the efficiency during the experimental results may be obtained by plotting the percentage of recovered CO_2 during the various runs. Fig. 4 shows the CO_2 recovery against the liquid flow at 8.2 m, 6.6 m, 4.9 m, and 3.3 m of absorption height.

For 8.2 m and 6.6 m absorption heights there is almost 100% removal efficiency. At these conditions little effect of liquid flow is observed, since the column is sufficiently high to reach almost complete equilibrium in the top outlet gas. The efficiency drops slightly to 98.8% at 8.2 m and 97.5% at 6.6 m at the low liquid flow of 2.1 L/min. The CO_2 recovery decreases both with decreasing column height and with decreasing liquid flow, as expected. However if the column is high enough, the liquid flow is less crucial. Similarly if the liquid flow is sufficiently high, the effect of column height is reduced. The reduction in CO_2 recovery is most apparent at low to moderate column heights where the liquid flow is also low. This

Table 2
Overview of experimental runs.

L/G	Absorption height/m				
	1.6	3.3	4.9	6.6	8.2
4	R20	R17	R11	R6	R1
5.5			R12	R7	R2
8		R18	R13–R14, R21–R23	R8	R3
11			R15	R9–R10	R4
18		R19	R16		R5

Table 3
Operation conditions.

Runs	Column height (m)	FI2 (L/min)	FI3 (g/min)	FI1 (L/min)	Ci1 (%)	Ci2 (%)	L/G (mol/mol)	Pressure external (mbar)	Temp. surrounding (°C)
R1	8.2	519	110.4	2.1	9.3	0.1	3.9	1005	22
R2	8.2	523	112.2	3.0	9.3	0.1	5.5	1005	22
R3	8.2	509	112.2	4.2	9.5	0.0	7.9	1017	23
R4	8.2	507	112.2	6.0	9.5	0.0	11.3	1005	22
R5	8.2	466	113.2	8.9	10.5	0.0	17.7	1000	21
R6	6.6	522	112.2	2.1	9.5	0.2	3.9	1005	22
R7	6.6	524	113.2	3.0	9.4	0.2	5.5	1005	22
R8	6.6	511	112.2	4.2	9.7	0.1	7.9	1017	23
R9	6.6	508	112.2	6.0	9.4	0.0	11.3	1005	22
R10	6.6	490	111.3	6.0	9.8	0.2	11.4	1000	21
R11	4.9	516	108.6	2.1	9.5	1.0	3.9	1000	21
R12	4.9	542	108.6	3.0	9.4	0.9	5.3	1000	21
R13	4.9	526	111.3	4.2	10.1	0.8	7.5	1000	21
R14	4.9	522	106.7	4.2	9.6	0.7	7.7	1017	23
R15	4.9	501	111.3	6.0	10.1	0.7	11.1	1000	21
R16	4.9	468	111.3	9.1	10.4	0.1	18.1	1000	21
R17	3.3	513	110.4	2.1	9.2	1.6	3.9	1000	21
R18	3.3	523	111.3	4.2	10.9	2.1	7.8	1017	23
R19	3.3	486	111.3	9.1	10.2	0.4	17.5	1000	21
R20	1.6	498	110.4	9.1	9.7	0.8	17.3	1000	21
R21	8.2	503	106.7	4.2	9.6	0.4	8.1	1000	21
R22	8.2	502	107.6	4.2	9.9	0.6	8.1	1000	21
R23	8.2	499	106.7	4.2	10.2	0.9	8.2	1000	21

trend is clear in the 4.9 m column, where the recovery decreases from 98 to 90%. There is a slight (unexpected) drop in recovery from 83% to 80% from run 17 to run 18 as the liquid flow increases from 2.1 to 4.2 L/min. The experimental uncertainty can be expected to increase at lower liquid flowrates, where liquid channeling may occur. This may not be as big a problem for taller columns with regular liquid redistribution, but will be exacerbated by lower column heights, where gas bypassing the liquid is more likely to take place. Theoretically the CO_2 removal efficiency should converge to zero when the liquid flow goes to 0 L/min, but this operating region is inaccessible since the flowmeter Fl1 has a lower flow limit of 2.1 L/min.

Fig. 4 shows that the liquid flow has greatest influence on the absorption efficiency when the absorption height is low. When the experiments are run at an absorption height of 6.6 or 8.2 m, where the absorption efficiency is almost independent of absorption height. At these column heights the recovery does increase, but only very slightly for a liquid flow rate above 3 L/min. In the range of 3–6 L/min the efficiency is almost constant at fixed columns heights, but for low flows (2–3 L/min) and high flows (6–9.1 L/min) the efficiency increases. Fig. 4 also reveals that a tall column has a similar effect to a high flow rate. In order to reach 95% recovery either a tall column is needed or (for a smaller column) a higher liquid flow. Regardless of the liquid flow, our exploration of the height/flow space reveals that a tall column (4.9 m or higher) is required to achieve 90% recovery. This naturally suggests operating at a high liquid flowrate (and therefore a larger diameter column). This combination is expected to reduce capital and operating costs in a larger (demonstration or full-scale) plant.

5.3. Experimental temperature profiles

Figs. 5 and 6 show the temperature profiles for selected runs at 8.2 m and 4.9 m absorption heights. The curves represent different

Table 4
Average temperatures in the system during steady state operation.

Runs	TI1 (°C)	TI2 (°C)	TI3 (°C)	TI4 (°C)	TI5 (°C)	TI6 (°C)	TI7 (°C)	TI8 (°C)	TI9 (°C)	TI10 (°C)	TI11 (°C)	TI31 (°C)	TI43 (°C)
R1	20.8	21.4	21.3	22.6	23.6	26.3	29.5	37.1	47.9	52.6	39.2	26.4	20.2
R2	20.7	21.0	20.8	21.8	22.5	24.2	26.3	29.7	35.2	45.7	39.7	26.7	19.9
R3	23.5	23.8	24.0	24.2	24.6	25.8	27.0	29.1	30.3	38.4	35.2	28.2	23.2
R4	20.3	20.3	20.1	20.5	20.8	21.7	22.8	24.2	25.8	30.1	30.8	26.9	19.6
R5	20.0	20.1	20.0	20.1	20.3	20.9	21.4	22.2	23.4	25.8	27.8	24.8	19.7
R6			20.8	21.7	23.2	25.6	28.9	35.0	42.9	49.2	38.8	27.3	20.3
R7			20.5	21.1	22.0	23.7	25.5	28.8	32.9	40.2	37.2	27.2	20.0
R8			23.7	24.2	24.5	25.8	27.1	29.1	29.6	38.2	35.0	28.2	23.1
R9			20.1	20.3	20.6	21.4	22.4	24.0	25.2	29.1	30.2	27.1	19.6
R10			19.9	20.2	20.6	21.5	22.4	24.0	25.7	30.3	30.0	24.2	19.6
R11					20.8	23.7	27.0	35.1	50.9	53.7	39.1	23.6	19.9
R12					22.6	23.0	25.2	28.4	34.7	44.1	40.4	26.0	19.9
R13					21.9	22.4	23.3	25.4	29.8	35.5	35.0	25.9	19.8
R14					23.6	25.2	26.5	28.6	29.6	38.2	36.4	26.7	23.1
R15					21.6	21.2	22.1	23.6	25.8	29.4	30.6	26.0	19.7
R16					20.9	20.5	21.1	21.8	23.0	25.3	27.6	25.0	19.5
R17							23.2	27.0	39.4	50.5	39.9	24.0	19.9
R18							27.0	26.9	27.9	37.3	35.8	28.5	23.1
R19							21.7	21.2	22.4	24.6	26.9	24.8	19.4
R20									24.0	22.5	24.9	24.5	19.4
R21	24.0	25.0	24.8	25.6	26.3	28.1	29.6	31.7	34.6	38.4	35.8	26.6	24.0
R22	26.1	27.4	27.1	28.1	29.0	31.1	32.7	34.9	37.8	41.2	37.6	27.2	26.3
R23	27.9	29.4	29.0	30.2	31.3	33.5	35.4	37.6	40.7	43.1	39.4	27.8	28.1

Table 5
Pressure profile and mass balances.

Runs	PI1 (mbar g)	PI6 (mbar g)	PI11 (mbar g)	Mass balance (%)
R1	13.8	17.8	19.6	11.0
R2	4.9	9.0	10.3	9.7
R3	10.6	10.4	9.5	−2.1
R4	3.3	8.1	9.8	1.8
R5	30.5	34.4	41.0	−0.3
R6	14.0	17.5	17.0	0.9
R7	7.4	11.3	10.9	5.5
R8	14.3	14.2	13.6	−4.4
R9	1.9	6.4	7.9	0.5
R10	19.5	24.0	30.3	−0.4
R11	9.1	12.6	18.1	5.9
R12	22.3	23.1	26.3	11.1
R13	39.5	40.4	43.6	−2.7
R14	10.4	11.4	12.9	−1.8
R15	40.3	40.6	43.1	−2.1
R16	24.9	28.1	35.6	−8.5
R17	17.2	20.5	25.4	1.0
R18	10.0	8.2	7.0	−4.2
R19	23.1	26.0	32.1	1.2
R20	15.2	18.7	24.1	3.8
R21	23.2	22.1	17.4	−4.9
R22	20.3	17.3	11.2	−2.6
R23	23.4	17.9	11.7	−3.5

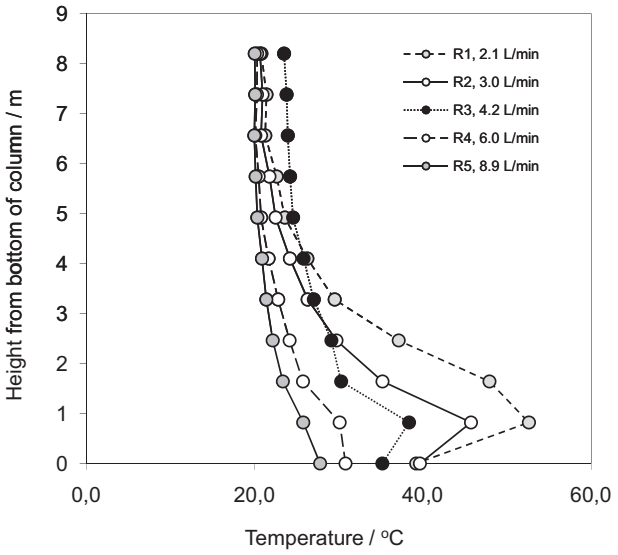


Fig. 5. Temperature profiles for a 8.2 m high column with a liquid flow from 2.1 to 9.1 L/min.

runs at various L/G ratios. In Fig. 5 it can be seen for all cases that the temperature barely changes in the topmost part of the column. The reason is that nearly all CO_2 is absorbed by this point so no exothermic reaction between solvent and CO_2 occurs. For a packed height of 4.9 m, shown in Fig. 6, most temperature profiles change all the way to the top, indicating that the full column height is required for absorption.

By comparing Figs. 5 and 6 it can be seen that the profiles from 0 m to 4.9 m in the 8.2 m column are similar to the profiles in the 4.9 m column. So increasing the column height from 4.9 m to 8.3 m has little effect, except where liquid flows are very low, for example in R11 (2.1 L/min) where only 89% recovery is achieved for the 4.9 m column. It should also be noted that the temperature bulge increases for decreased liquid flows. On the other hand this bulge disappears when the liquid flow is high enough (above a threshold value of about 6 L/min). It can be concluded that the liquid flow rate has a clear effect on the temperature profiles. It was previously determined that the temperature bulge does indeed have an effect on the absorption rates in the column (Dugas, 2006). The reason for this is that both the kinetics and the equilibrium of the absorption reaction depend on temperature to an extent sufficient for it to affect the absorption rate over the range of temperatures found in the column. In order to obtain more detailed understanding of the sensitivity towards height and liquid flow, two cases are considered with very different experimental conditions i.e. runs R11 and R5 discussed below.

5.4. Comparison of experimental and modelling temperature profiles

Fig. 7 shows the model-predicted temperature profiles in the column for runs R5 (column height 8.2 m) and R11 (column height 4.9 m). Run R5 is a case of high L/G ratio (17.7 mol/mol) for a tall

Table 6
Liquid CO_2 content mol_{CO_2}/mol_{MEA} at given positions in the column.

Runs	V0 Tank	V1 8.2 m	V2 7.38 m	V3 6.56 m	V4 5.74 m	V5 4.92 m	V6 4.1 m	V7 3.28 m	V8 2.46 m	V9 1.64 m	V10 0.82 m
R1	0.112										0.356
R2	0.112										0.286
R3	0.112	0.113	0.115	0.117	0.120	0.127	0.132	0.141	0.159	0.215	0.214
R4	0.112										0.190
R5	0.112	0.115	0.117	0.115		0.122	0.117	0.118	0.136	0.145	0.165
R6	0.112										0.325
R7	0.112										0.275
R8	0.112			0.115	0.117	0.132	0.131	0.143	0.171	0.217	0.211
R9	0.112										0.187
R10	0.112								0.151	0.178	0.187
R11	0.112					0.125	0.133	0.160	0.186	0.264	0.325
R12	0.112								0.175	0.254	0.288
R13	0.112								0.161	0.207	0.217
R14	0.112					0.118	0.124	0.132	0.156	0.206	0.212
R15	0.112								0.147	0.176	0.184
R16	0.112										0.149
R17	0.112								0.172	0.211	0.291
R18	0.112							0.125	0.149	0.207	0.206
R19	0.112										0.165
R20	0.112										0.165
R21	0.254	0.252			0.247		0.274		0.297	0.341	0.338
R22	0.271		0.281		0.293		0.299		0.334	0.364	0.364
R23	0.300		0.314		0.323		0.336		0.364	0.387	0.387

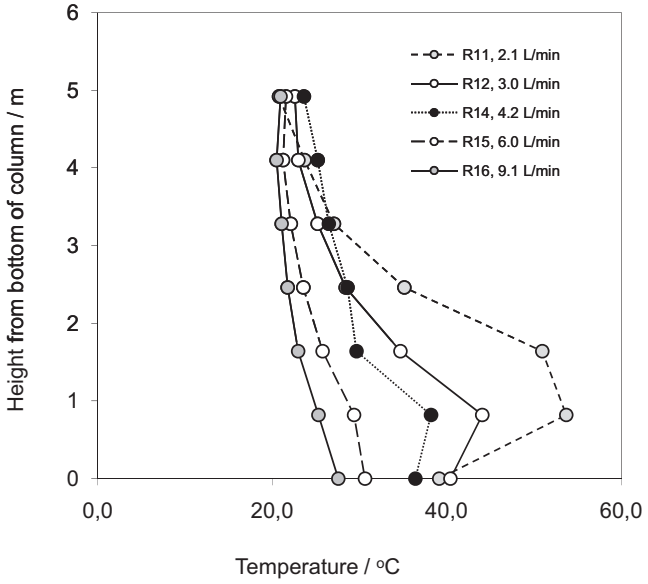


Fig. 6. Temperature profiles for a 4.9 m high column with a liquid flows from 2.1 to 9.1 L/min.

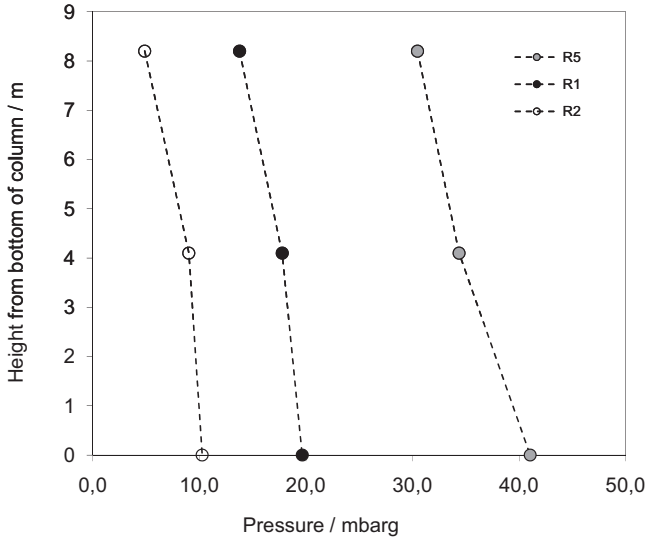


Fig. 8. Pressure profiles for 3 selected experiments. Three pressures are measured by gauges located at the top, middle and bottom of an 8.3 m column.

column and run R11 is a case with a low L/G ratio (3.9 mol/mol) and smaller column height.

For run R5 the temperature difference between inlet and outlet is low, around 8 °C. This is due to the high liquid flow rate (9.1 L/min). Good agreement between the modelling and experiment is observed. A small deviation (approximately 1 °C) is noted in the temperature prediction at the bottom. Run R11 shows a greater deviation between the experimental values and the model predictions (around 5 °C at the bulge). However the error is not that significant when it is taken into account that the temperature across the length of the column varies much more in this case – around 18 °C difference from top to bottom. The temperature bulge may be located in the bottom of the column as shown in Fig. 7 or it may exist as a large bulge throughout the column. An intermediate region where the bulge lies between runs R5 and R11 (Fig. 7) is difficult to obtain. In an extremely narrow range of L/G the bulge may be located in the middle of the column. This range could not be found

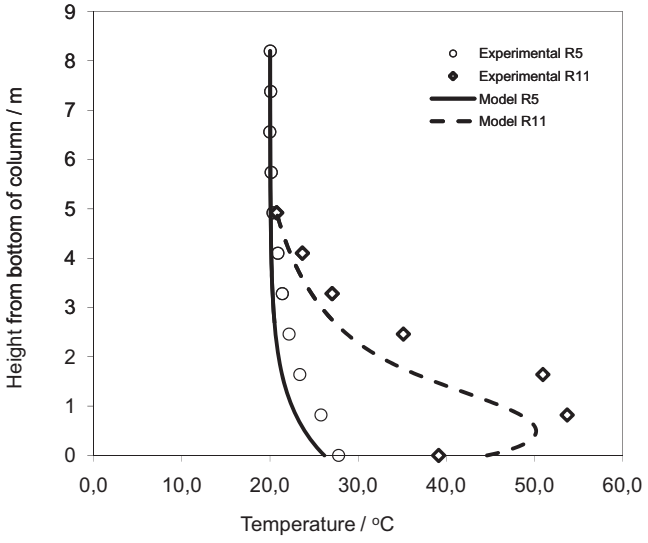


Fig. 7. Experimental results and modelling. R5 is absorption height of 8.2 m and a liquid flow of 9.1 L/min, while R11 is an absorption height of 6.6 m and a liquid flow of 2.1 L/min.

in this study. The column is said to be "ignited" under these conditions. Quantitative agreement between model and experiment is difficult to obtain in this case, since the "ignited" condition is very sensitive to process conditions. This means it is difficult to predict ignition at the exact same process conditions as the experiment. The temperature profiles will completely change shape in this narrow operating range from concave to convex. The model is better able to reproduce the experimental conditions at column conditions far from the ignited state. While minor deviations between model and experimental values may occur, in the middle section of the column, the model still predicts the outlet temperature with good accuracy.

5.5. Pressure profiles

The column pressure drop measured in this work is shown in Table 5. A number of factors contribute to it. First of all it is expected that the packing material will be a source of some pressure drop, but also the fan P32 in Fig. 2 will create a pressure drop due to the fact that it is not completely air tight. It is clear from Fig. 8 that there are some differences in pressure drop between experiments. Run R5 has a higher pressure drop which is evenly distributed throughout the column. However for run R1 and R2 the main drop occurs in the top part of the column. The main reason for that is, that the fan causes suction in the top of the column creating a slight positive pressure at the bottom of the column. Much of the pressure drop can therefore be prevented by not recycling the inert gas. While some pressure drop is observed, it can reasonably be neglected as it is fairly constant in all the experiments. There is a very slight difference (maximum −30 mbar) in pressure at the bottom of the column from run to run. This is a result of the different gas and liquid flow profiles for each run.

5.6. Modelling CO_2-loading profiles

The CO_2-loading profiles are affected by the liquid flow and the column height. This is illustrated for R5 and R11 in Fig. 9 which clearly shows that there is a big difference in the way the column performs under varying conditions. For R5 nearly all CO_2 is absorbed in the bottom part of the column leaving the top part ineffective. For R11 the change in loading has a much bigger effect (4 times greater). Here it can be seen that the entire column is needed

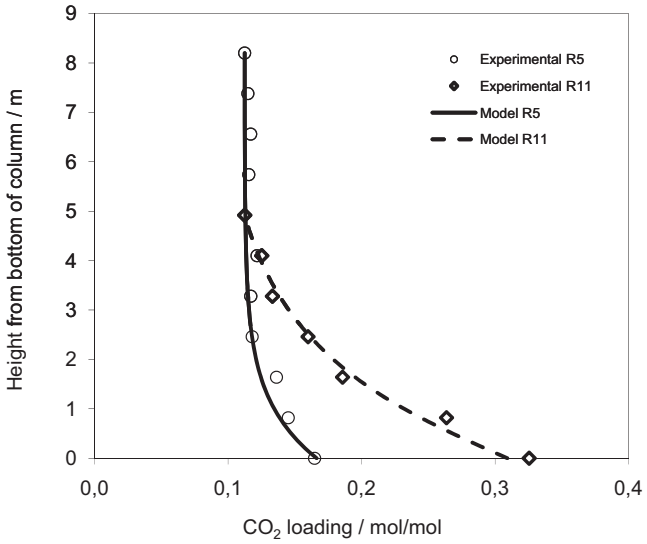


Fig. 9. Loading profile as a function of height for R5 and R11. The liquid flow is respectively 9.1 L/min and 2.1 L/min.

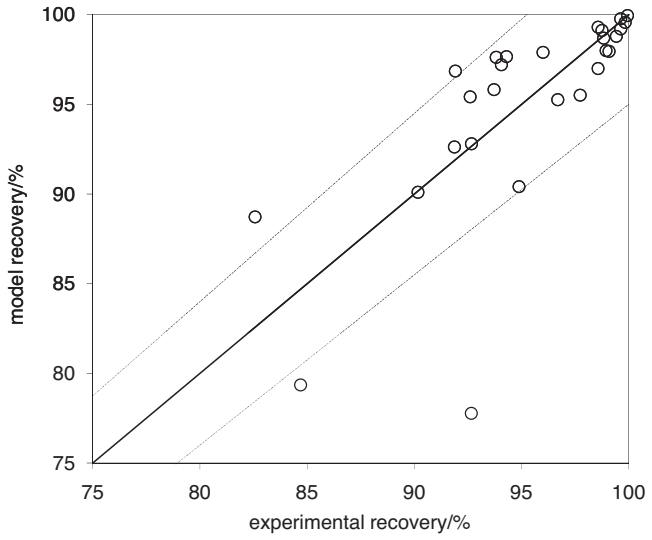


Fig. 11. Parity plot of CO_2-recovery. ±5% error lines are added to the plot.

in order to absorb all the CO_2. In both cases the model is able to predict the development of the loading profiles with good accuracy. In particular, it accurately predicts the bottom outlet concentration which is important for the model estimation of the rich solvent composition. For R11 the recovery is 90% and in R5 it is 99.9%. This means that in order to remove the very last 10% CO_2 from the flue gas it will require more absorption height or more liquid flow. It will undoubtedly increase the capital cost or operation cost in a full scale plant.

5.7. Modelling gas concentration profiles and recovery

CO_2 gas concentration profiles were measured at only two places in the pilot plant: At the inlet and at the outlet. Although only two experimental points are shown (inlet and outlet) the model predictions provide insight into how the CO_2 is absorbed as the flue gas rises in the column. A comparison between modelling results and experimental data is shown in Fig. 10, where a very low CO_2 concentration is predicted in the gas outlet. The relative change in concentration is highest in the bottom part of the column from 0 to 5 m. Similarly the driving force for transporting CO_2 from the gas phase to the liquid phase is also greatest in this part of the column. The model is capable of predicting the top outlet composition of the cleaned gas. This is an important process parameter since this is the gas which is sent to atmosphere. The CO_2 recovery may be evaluated based on the calculated gas concentrations. Fig. 11 is a parity plot, comparing simulations and experimental recoveries for all runs. The difference between model predictions and experimental values is less than 5% (based on experimental values) for all but 3 of the runs exceed this limit – the cases of very low absorption heights. Here the profiles become very sensitive towards the performance of the model and the CO_2-recovery.

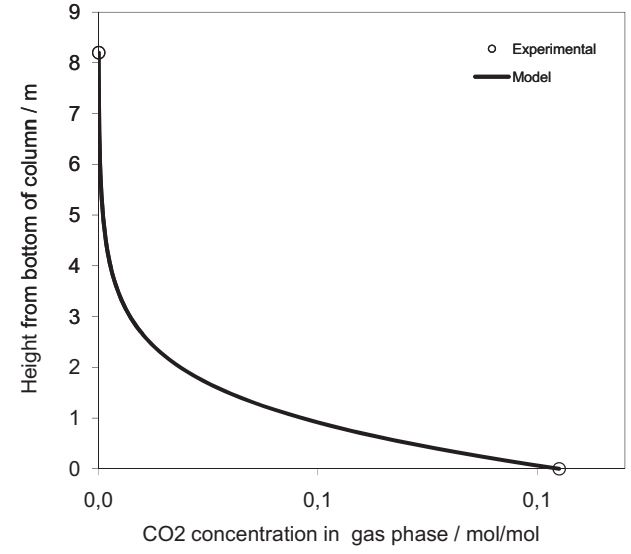


Fig. 10. The gas concentration for R5 with a liquid flow rate of 8.9 L/min plotted together with the model simulation. Gas was only sampled at the inlet and outlet of the column.

5.8. Variation of lean loading

In order to simulate a case of varying CO_2 lean loading in the return line from the stripper was varied between 0.11 and 0.30 mol/mol while all other conditions were kept constant. The CO_2-recovery decreases from 99.8% to 92% during this process as shown in Fig. 12. At high lean loading a higher amount of CO_2 is present in the liquid. Consequently the solvent cannot absorb as much CO_2 and the driving force is diminished. A decrease in recovery with higher loading is therefore expected.

5.9. Sources of error during modelling

To minimize errors all instruments were calibrated prior to each experimental campaign. It is believed that the manual flow meters are the sources of most deviation. Accuracy in flow measurement could be improved by replacing the manual instruments with digital readouts which are also logged directly to the computer.

Adiabatic conditions are assumed in the modelling although this is difficult to obtain in lab experiments. This may be the cause of some deviations in the temperature profiles between the model and the experimental work. A rough conservative estimate of the amount of heat that is escaping the system may be obtained by neglecting film resistance on walls and assuming the bulk temperature extends to the internal column walls and that the external walls are at room temperature. The column walls consist of 1.3 cm thick glass. This gives a maximum 0.1 °C drop in temperature for a

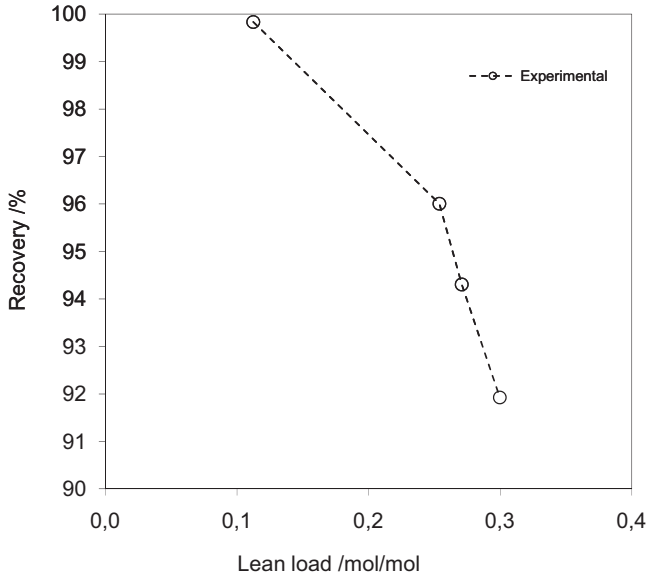


Fig. 12. The recovery as a function of lean loading varying from 0.11 to 0.30 mol/mol.

30 wt% MEA solution. It is thus clear that adiabatic conditions are a reasonable assumption for this system.

6. Conclusion

Experimental data (CO_2 recovery, temperature profiles, pressure profiles, CO_2 loadings and CO_2 gas phase concentrations) for a pilot plant absorber using structured packing (Mellapak 250Y) for capturing CO_2 with 30 wt% MEA are presented and validated against an in-house absorber model.

There is good agreement between the experimental values and model predictions with respect to temperature, CO_2-loading, and CO_2 gas phase concentration. Some deviations are observed where the model predicts the onset of a so-called "ignited" state. An ignited state was not observed experimentally.

A temperature bulge was observed which moved upwards and changed size as various L/G-ratios were applied. A bulge was never observed in the top part of the absorber."

Appendix A. Analysis method for CO_2 content

- Take a 100 mL Erlenmeyer flask and dispense 5 mL 0.1 M NaOH from the automatic dispenser.
- Weigh the flask and tare the scale.
- Use a pipette and dispense:
  - 1 g of sample into the flask if there is high load capacity.
  - 1.5 g of sample into the flask if there is low load capacity.

  To determine whether the sample has a low or high loading capacity a test can be made by running the analysis with a sample of 1 g and if the loading is less than 0.15 mol_{CO_2}/mol_{MEA}, then the concentration is too low and the weight of the sample should be increased to 1.5 g for better precision.
- Then dispense 25 mL 0.5 M $BaCl_2$.
- Place the flask on the warming plate and bring it to boil. After boiling for 4 min cool it down again.
- When mixture is cooled down to about room temperature, it is transported to the vacuum filtration apparatus. The filter should be a 45 μm cellulose nitrate filter.
- Start vacuum pump and wet the filter with a little bit of demineralized.
- Put the cooled solution into the filter.
- Take four times 25 mL demineralized to get the last drop of solution from the flask and wash the filter with the water.
- After filtering, carefully take the filter with compound and place it again in the same Erlenmeyer flask as before. Add to the flask and filter 20 mL pure water to rinse the filter.
- Disperse with 12 mL 0.5 M HCl from a burette. Write down the exact volume as it has to be used for calculating the loading. CO_2 will then bubble off.
- Place the flask on the warming plate and bring it to boil and cool down again to room temperature. The heating will avoid formation of H_2CO_3.
- When the solution is at room temperature, put a magnetic stirrer in the flask and place the flask on the autotitrator.
- Calibrate the electrode with a buffer of pH 4.0 and 7.0.
- Titrate the solution with 0.100 M NaOH until pH 7. The pH should be stable for a period of time.
- Rinse the electrode and NaOH dispenser with demineralized water. Carefully dry it with a tissue.
- Place the flask on a magnetic stirrer and lower the electrode and dispenser into the solution.
- Start the titration program and after ending record the final amount of NaOH.
- The CO_2-loading can now be calculated:

$$\text{wt\% CO}_2 = \frac{C_{HCl} \cdot V_{HCl} - C_{NaOH} \cdot V_{NaOH}}{2 \cdot m_{sample}} M_{CO_2} \cdot 100$$

$$\text{CO}_2\text{-loading} = \frac{\text{wt\% CO}_2}{\text{wt\% MEA}} \cdot \frac{M_{MEA}}{M_{CO_2}}$$

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